(DEMO) From Natural Language to Live 5G API Calls: A CAPIF-Compliant, Multi-LLM Intent Orchestration Platform

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***Abstract*—We demonstrate an open-source orchestration platform that converts free-text operator intents into live CAMARA API calls whose Quality of Service (QoS) is authorized by a 3GPP PCF. A two-phase large-language-model (LLM) planner selects the CAMARA service and builds a schema-grounded payload, and a three-layer deterministic validator (schema, telecom semantics, live CAPIF registry) gates every call before it reaches a CAPIF-registered provider. The southbound path runs on a containerized Open5GS 5G core with UERANSIM-emulated UEs: a QoD request traverses CAMARA → NEF (TS 29.122) → PCF (TS 29.514) and is bound to the UE's PDU session. In recorded live runs a request completes in 2.2–4.8 s with GPT-4o-mini or Claude Haiku 4.5 (6.9–8.2 s with Gemini), 78–99% of it LLM planning. On 300 unique bilingual intents, the LLM planners reach an Intent-to-API Workflow Success Rate (IWSR) of 95.3–97.3%, versus 83.3% for a keyword baseline. Code, testbed, datasets and live runs: https://github.com/doadurak/mini-telco-platform.**

***Index Terms*—CAPIF, CAMARA, intent-based networking, LLM orchestration, 5G API exposure, Open5GS**

I. Introduction

Zero-touch Beyond-5G management requires orchestration layers that translate high-level intents into standards-compliant API invocations [1]. 3GPP CAPIF [2] and GSMA CAMARA [3] standardize how verticals discover and consume network capabilities, but closing the intent-to-API gap on a *live* exposure stack remains an open challenge. OpenCAPIF [4] provides exposure infrastructure without NL planning; Turkovic et al. [5] ran live CAMARA QoD with hand-written payloads; Maatouk et al. [6] identify hallucination as the key risk of telecom LLMs, and Liu et al. [7] report 40% JSON-schema compliance for ungrounded GPT-4. Our open platform combines live CAPIF onboarding, schema-grounded multi-LLM planning, a three-layer validator, and CAMARA execution down to a 3GPP PCF, reproducible from a single script.

II. Platform Architecture

Planning: A FastAPI backend exposes `POST /orchestrate` (Fig. 1). Phase 1 of the LLM planner selects one of four CAMARA services (QoD v1.1, QoS Profiles, QoS Provisioning, Location Retrieval v0.5) from a schema-grounded catalog; Phase 2 populates the payload using the operation's OpenAPI schema, which is injected [8]. `POST /config/provider` switches the LLM backend at runtime. Intent-level QoS classes (e.g. `QOS_E_STREAMING`) are mapped by target bit rate to the operator profiles the network enforces (`QOS_E/S/M/L`).

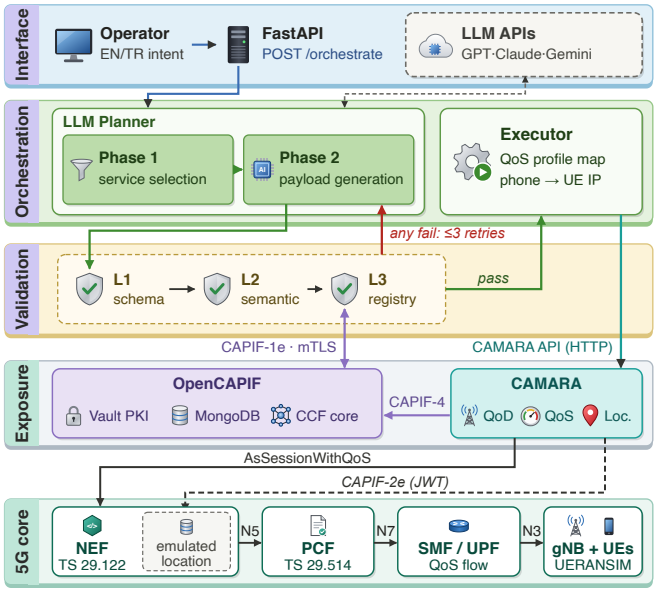


Fig. 1. Platform architecture. Any L1–L3 failure triggers at most three payload corrections; CAPIF-1e runs over mTLS, provider calls over HTTP.

Validation: L1 checks the payload against its Pydantic schema; its pass rate is the Schema Compliance Rate (SCR). L2 enforces telecom rules (E.164 phone numbers, QoS profiles that map to an operator profile, session durations of 1–7200 s, unicast IPv4 addresses); its pass rate is the Semantic Validity Rate (SVR). L3 queries the live CAPIF registry to confirm that the selected service is published. This catches non-existent services but not a wrong-but-registered one, which shows up as an IWSR miss against ground truth.

Execution: For QoS, the adapter resolves the phone number to the UE's IP address from the 5GC subscriber database,

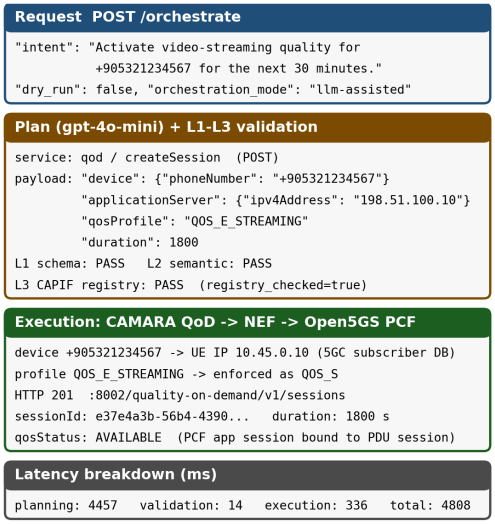


Fig. 2. Recorded live response (S1, GPT-4o-mini, `dry_run:false`), rendered from the stored JSON in the repository.

the CAMARA QoD provider forwards an AsSessionWithQoS subscription to the NEF, and the NEF creates a PCF application session (Npcf_PolicyAuthorization) bound to the UE's PDU session. Location uses the NEF Monitoring Event API with emulated core data.

III. Live Demonstration

The full stack (OpenCAPIF, four CAMARA/NEF providers, Open5GS, and a UERANSIM gNB with three UEs) runs on a single 16 GB host via `start.sh`. All scenarios run with `dry_run:false` and are versioned in the repository. **S1 (QoD):** *"Activate video-streaming quality for ⟨MSISDN⟩ for the next 30 minutes"* yields `QOS_E_STREAMING`, 1800 s, enforced as `QOS_S`; the PCF accepts the application session and QoD returns HTTP 201, `AVAILABLE` (Fig. 2). **S2 (Location):** the NEF returns a WGS-84 area with HTTP 200. **S3 (ambiguity):** dataset entry *"Provide network quality improvement for device ⟨MSISDN⟩ for a long time"* (ground truth: QoD). In the benchmark GPT-4o-mini chose QoD and Claude and Gemini QoS Provisioning; in the recorded live run all three chose QoS Provisioning (PCF: `ACTIVE`). Every plan passes L1–L3 and executes: the validator guarantees well-formed, registered calls, not intent correctness. **S4 (backend switch):** re-running S1 after each switch yields identical CAMARA payloads from all three LLMs. Across the nine runs, validation takes 13–27 ms and execution 0.03–0.6 s. UERANSIM does not implement NGAP PDU Session Resource Modify, so the dedicated QoS flow is authorized and signaled by the core, but not enforced on the emulated radio. *External LLM APIs:* the only external dependency; as the platform's own stages take milliseconds, their latency and rate limits bound throughput, and in `auto` mode an LLM error falls back to the keyword planner. Intents, including MSISDNs, leave the testbed, so production use would need pseudonymization or a self-hosted model.

TABLE I
300-ENTRY OFFLINE EVALUATION (ONE RUN PER INTENT)

Planner	IWSR	SCR	SVR	HR	SS
Keyword baseline	83.3%	100.0%	100.0%	16.7%	0.962
GPT-4o-mini	**97.3%**	100.0%	99.0%	**2.3%**	**0.975**
Claude Haiku 4.5	95.3%	99.3%	**99.3%**	4.0%	0.927
Gemini 2.5 Flash	96.0%	100.0%	98.3%	3.0%	0.945

IV. Evaluation

The dataset contains 300 unique intents (152 EN / 148 TR; 96 easy, 94 medium, 110 hard): 275 across the four services and 25 ambiguous edge cases. We derived the ground truth (service, operation, payload checks) from the CAMARA OpenAPI constraints and NEF test fixtures; it supersedes a 60-entry pilot kept in `datasets/legacy_pilot_60/`.[1] IWSR counts plans with the correct service that pass validation. HR is the share of incorrect service selections. SS is $1-\sigma$ of IWSR across the three difficulty strata (robustness to difficulty, not run-to-run repeatability). Planning uses temperature 0.1 and one run per intent, so choices on ambiguous intents can still vary between runs (cf. S3). The benchmark makes no provider calls; we revalidated all 1,200 recorded plans against the live CAPIF registry, and L3 rejected none. The keyword rules were written against the pilot, where the baseline scores 100%; on the 240 held-out entries it drops to 79.2% (LLMs: 95.4–97.1%), all 50 failures being wrong-service selections. The 8–14 LLM failures per model consist of 7–12 wrong-service selections (mostly QoD vs. QoS Provisioning, as in S3) and 1–3 L1/L2 rejections; hard intents account for most of them (hard IWSR 87–95% vs. 99–100% for easy and medium). The feedback loop ran for 5–7 plans per model and repaired 2–3 of them (RRF 29–60%): it fixes payload errors but cannot correct a wrong-but-valid service choice.

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[1]The first 60 LLM results per model are reused from the pilot; the Claude pilot predates three bug fixes, and a symptom scan found no affected entry. Eleven multi-intent entries that failed because Phase 1 returned a list of plans were re-run after adding multi-step support (flagged in the result files).